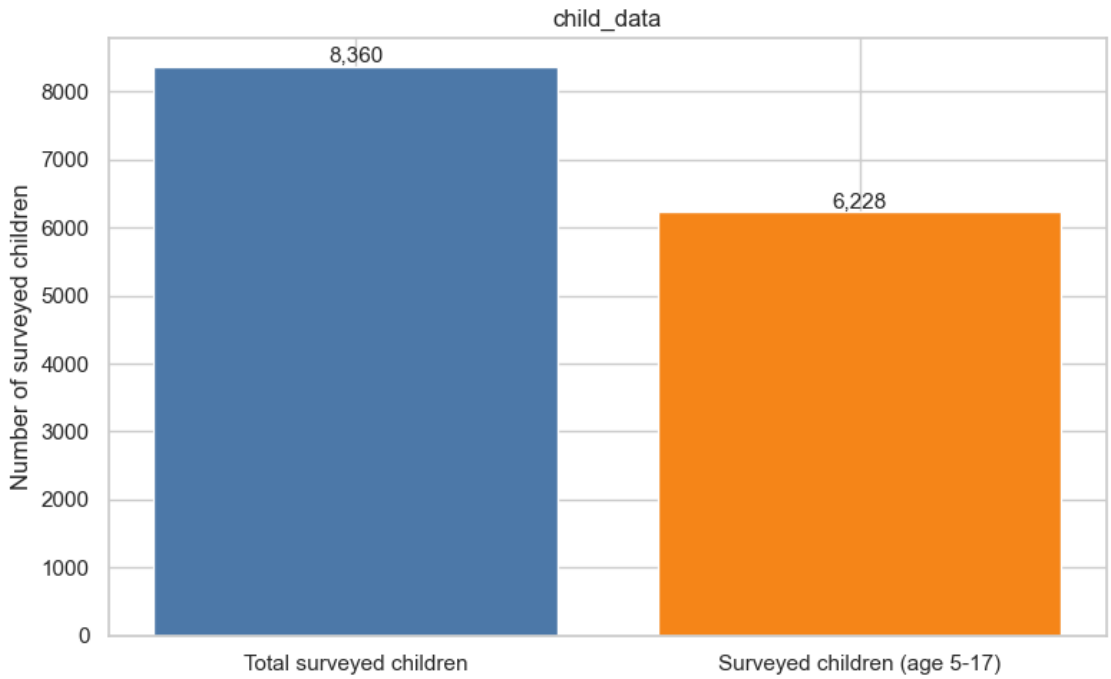
Child_Food_Allergy

April 1, 2026

0.1 Exploratory Data Analysis (EDA)

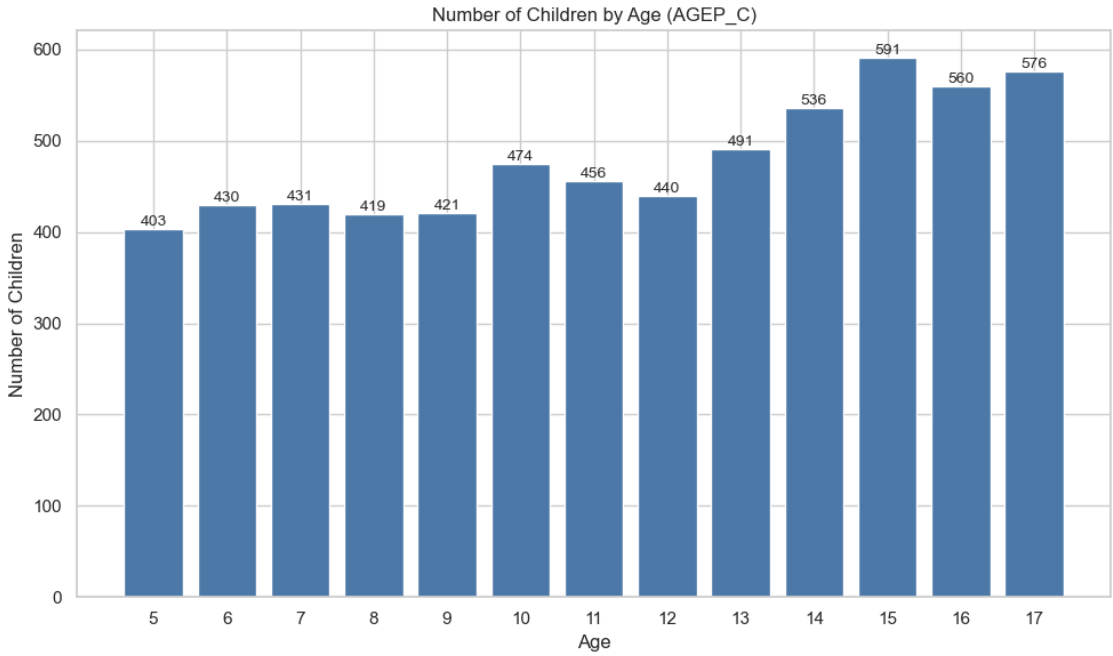
0.1.1 Sample Overview

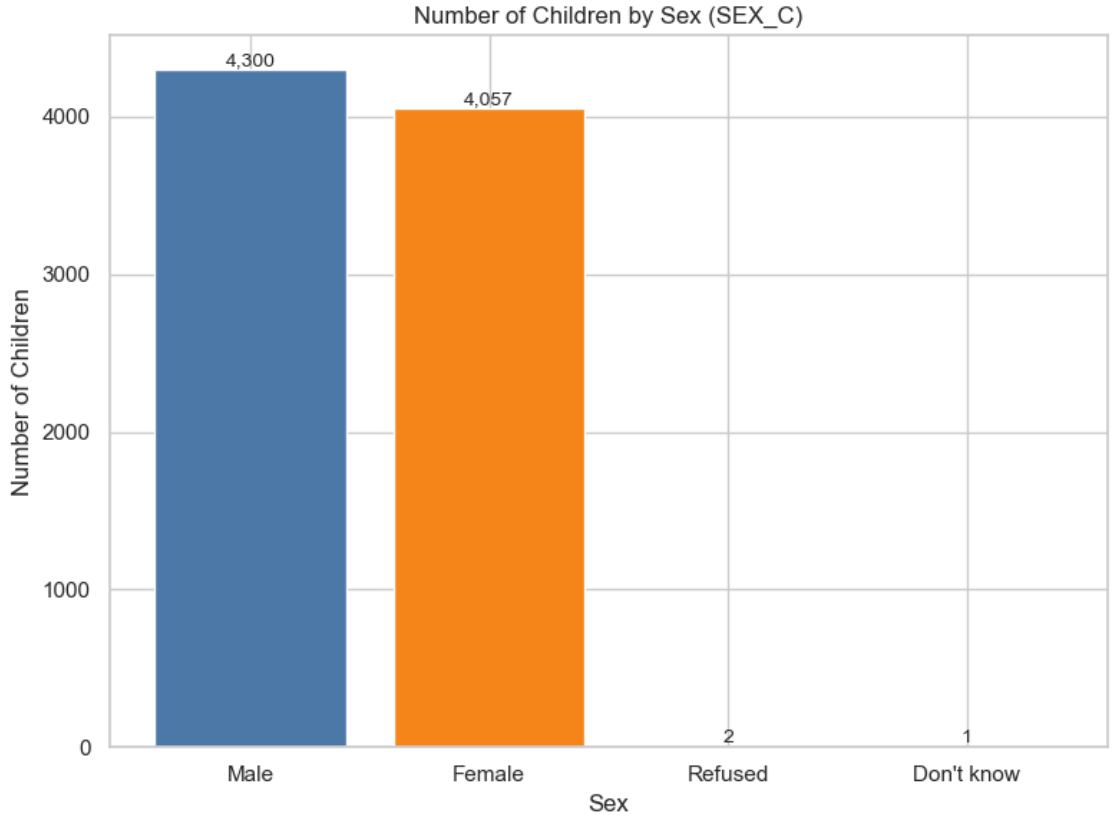
- Total number of surveyed children
- The number of surveyed children in selected age range 5 to 17

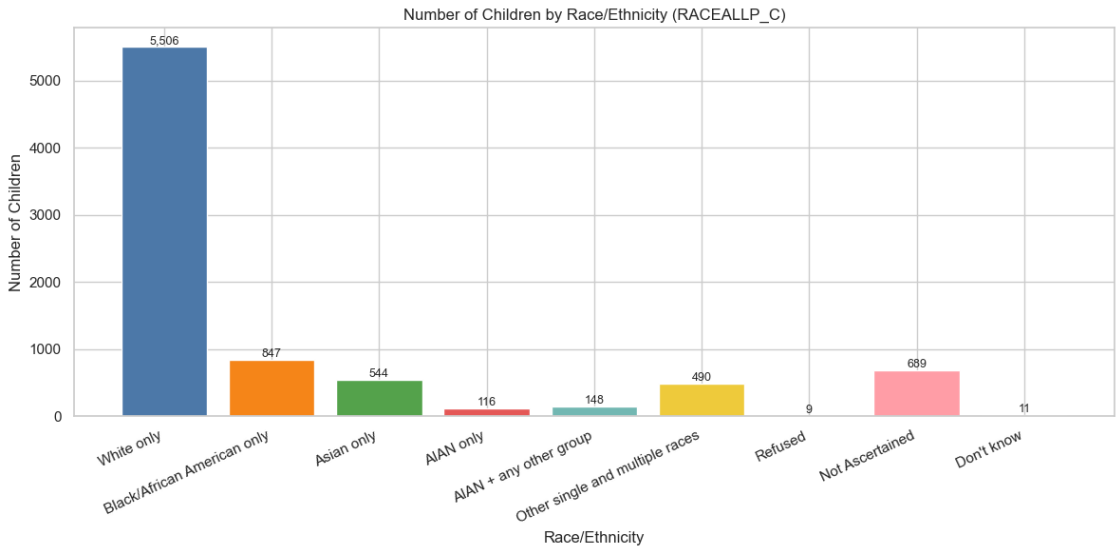


0.1.2 Demographics

- The number of children by age
- The number of children by sex
- The number of children by race/ethnicity

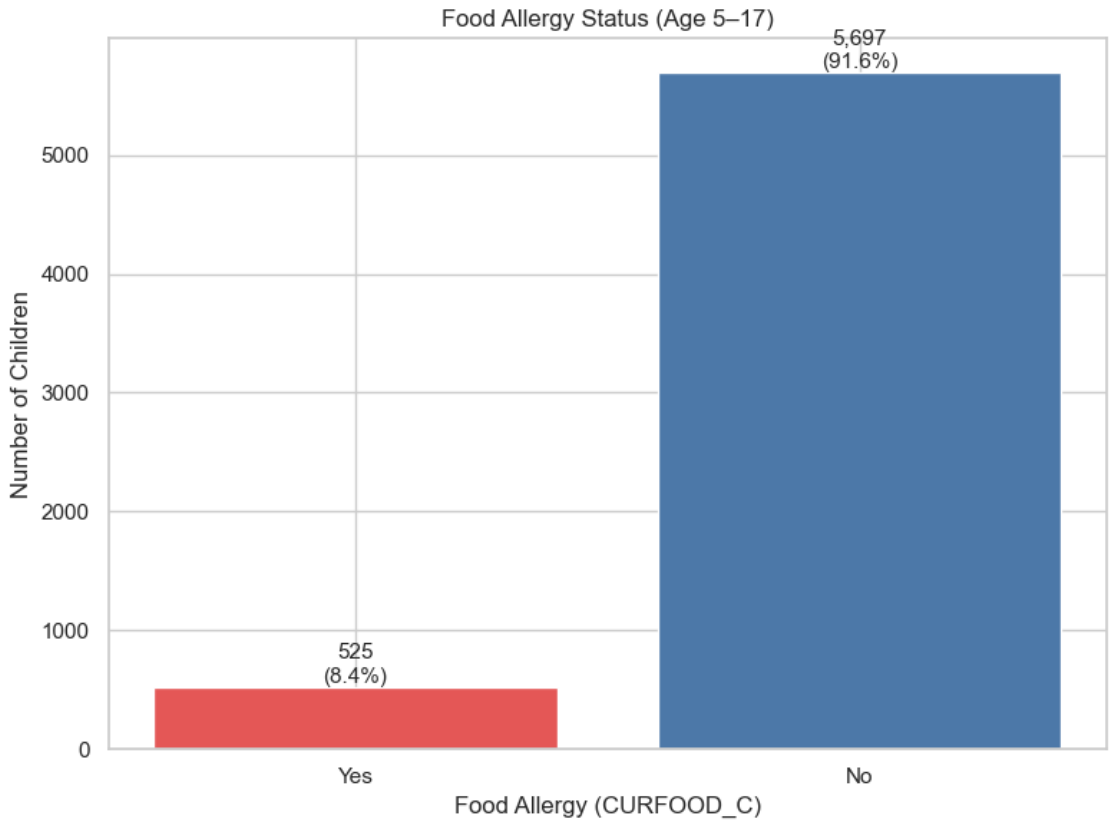

Number of Children by Age (AGEP_C)
Number of Children
Age
0
100
200
300
400
500
600
5
6
7
8
9
10
11
12
13
14
15
16
17
403
430
431
419
421
474
456
440
491
536
591
560
576


Number of Children by Sex (SEX_C)
Number of Children
Sex
0
1000
2000
3000
4000
Male
Female
Refused
Don't know
4,300
4,057
2
1

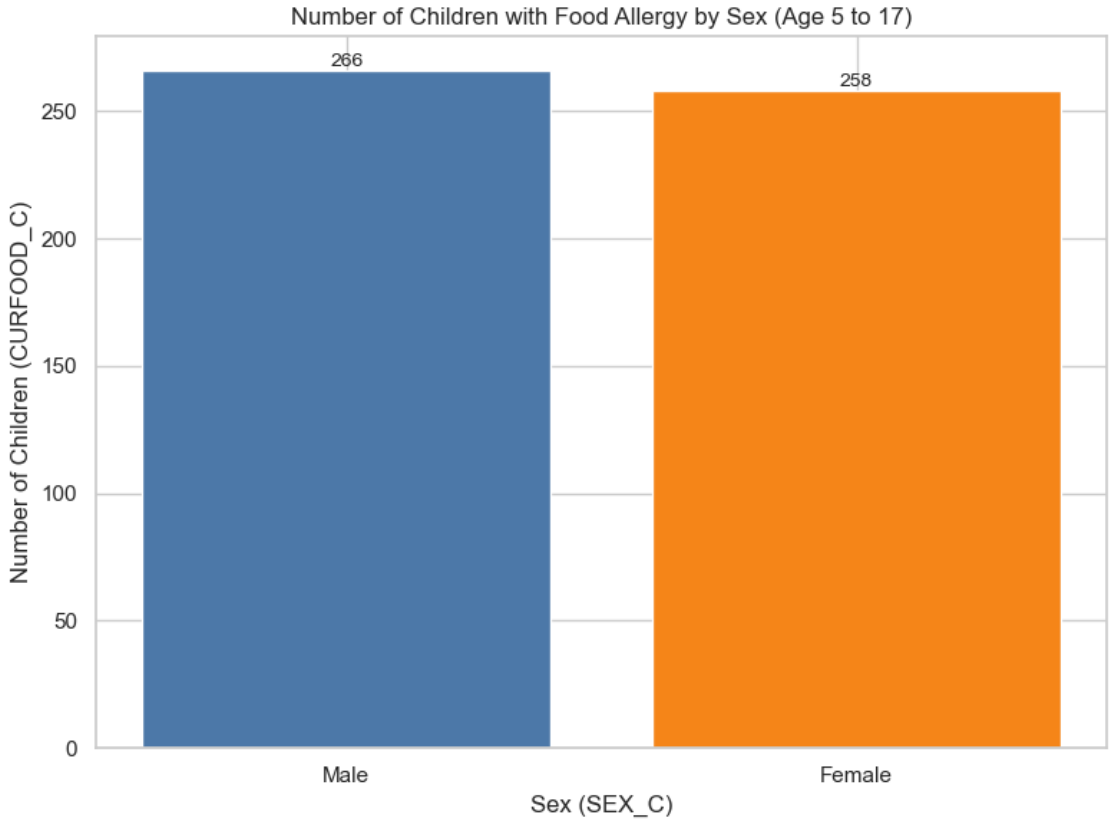


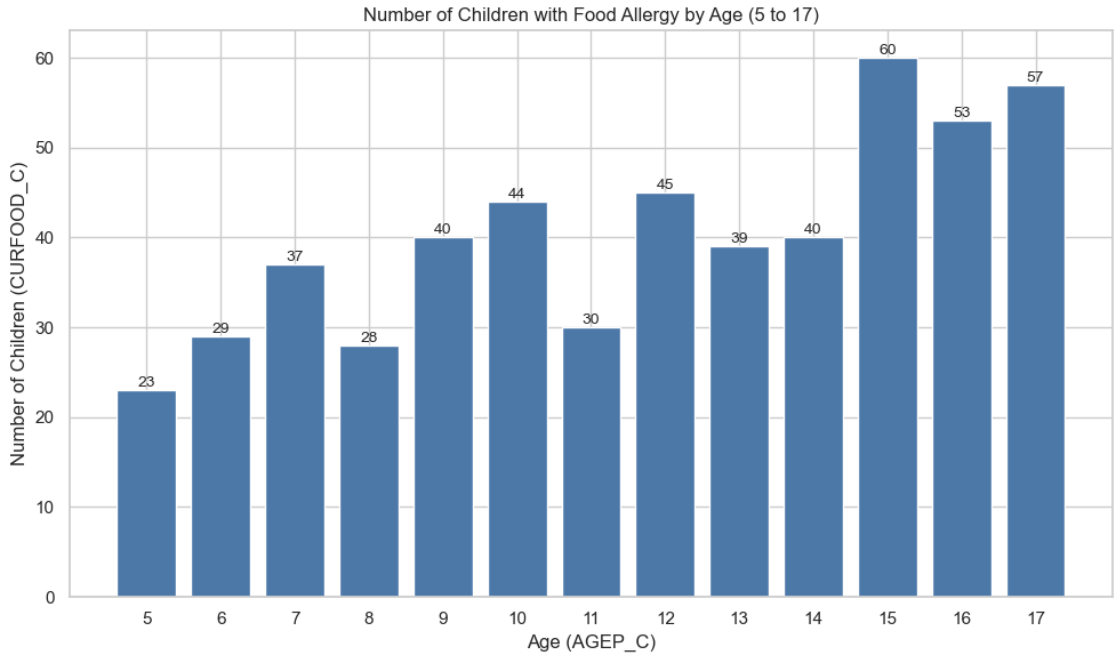
0.1.3 Food Allergy

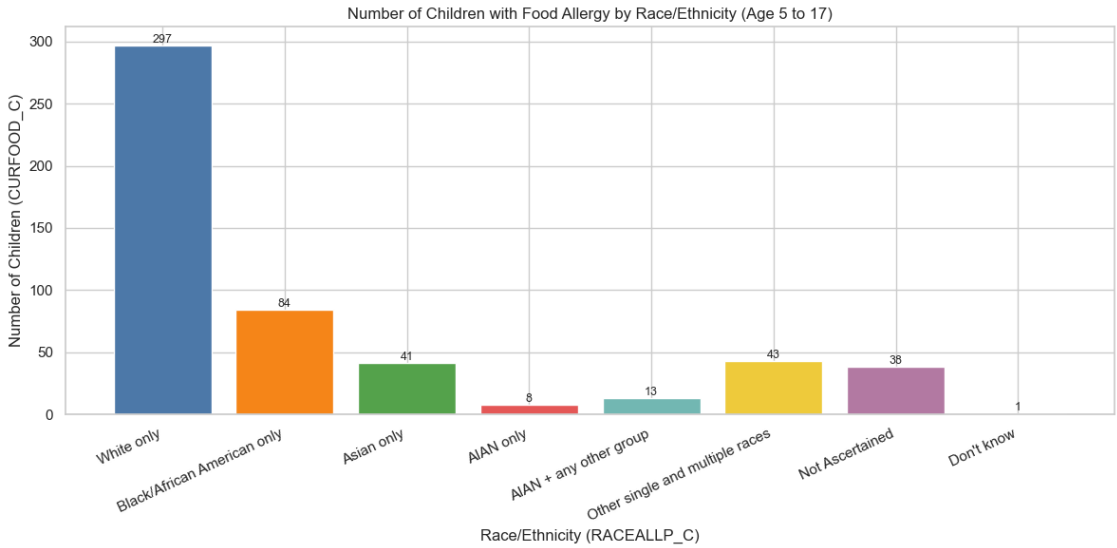
- The number of children with current food allergy (CURFOOD_C)
- Percentage of children with current food allergy



- The number of children having food allergy by sex
- The number of children having food allergy by age
- The number of children having food allergy by race/ethnicity

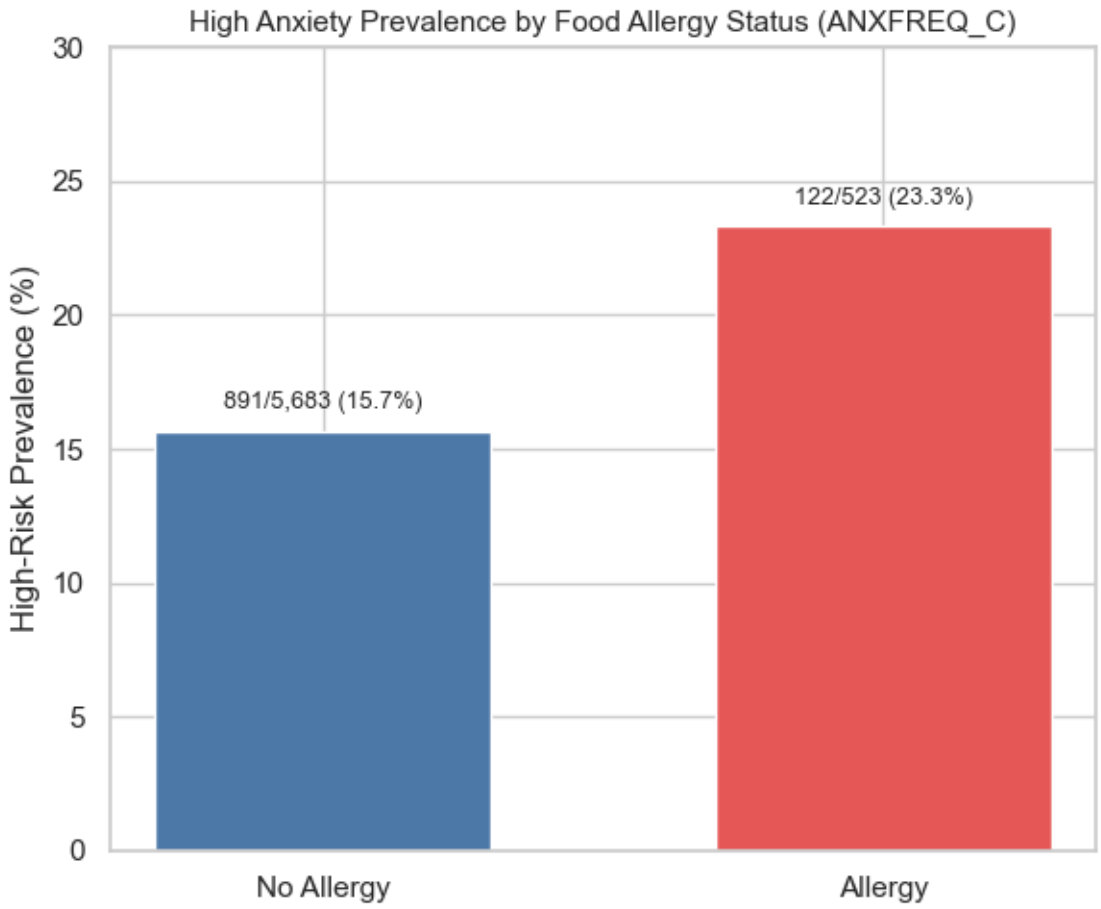


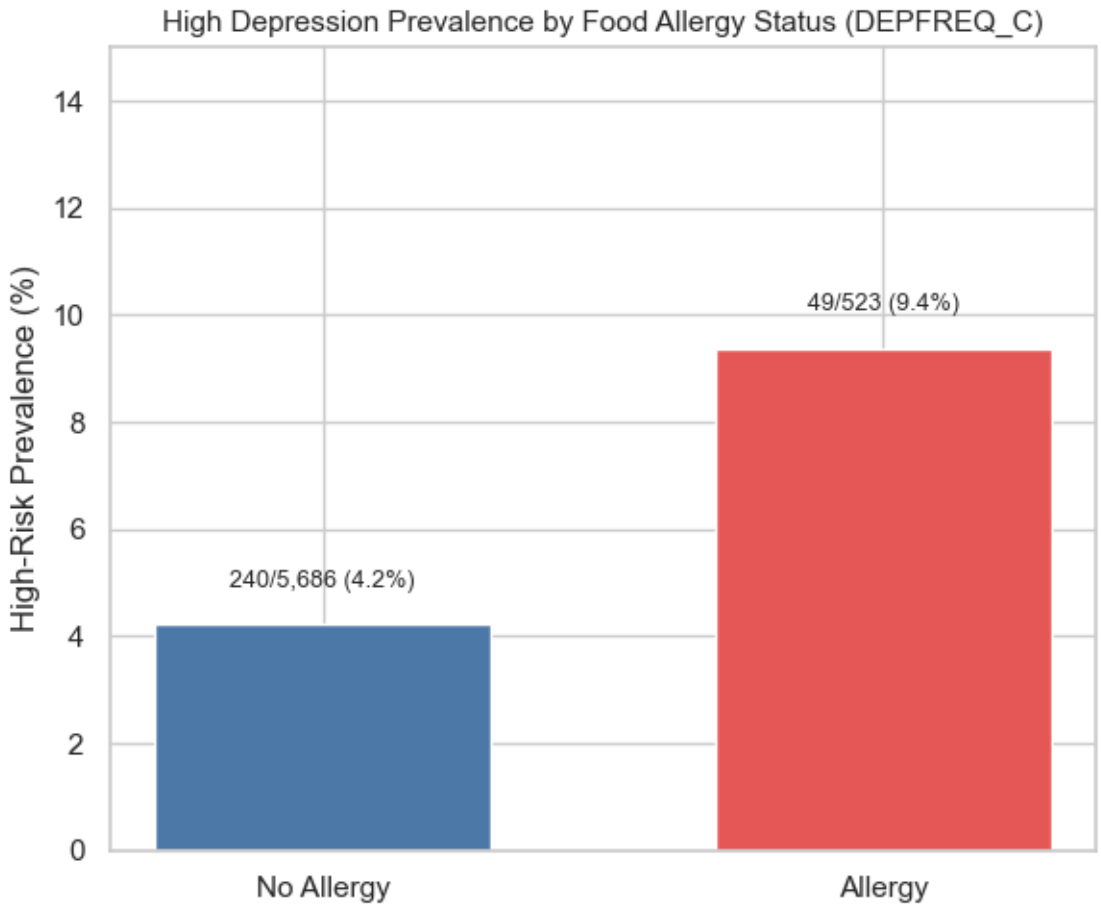


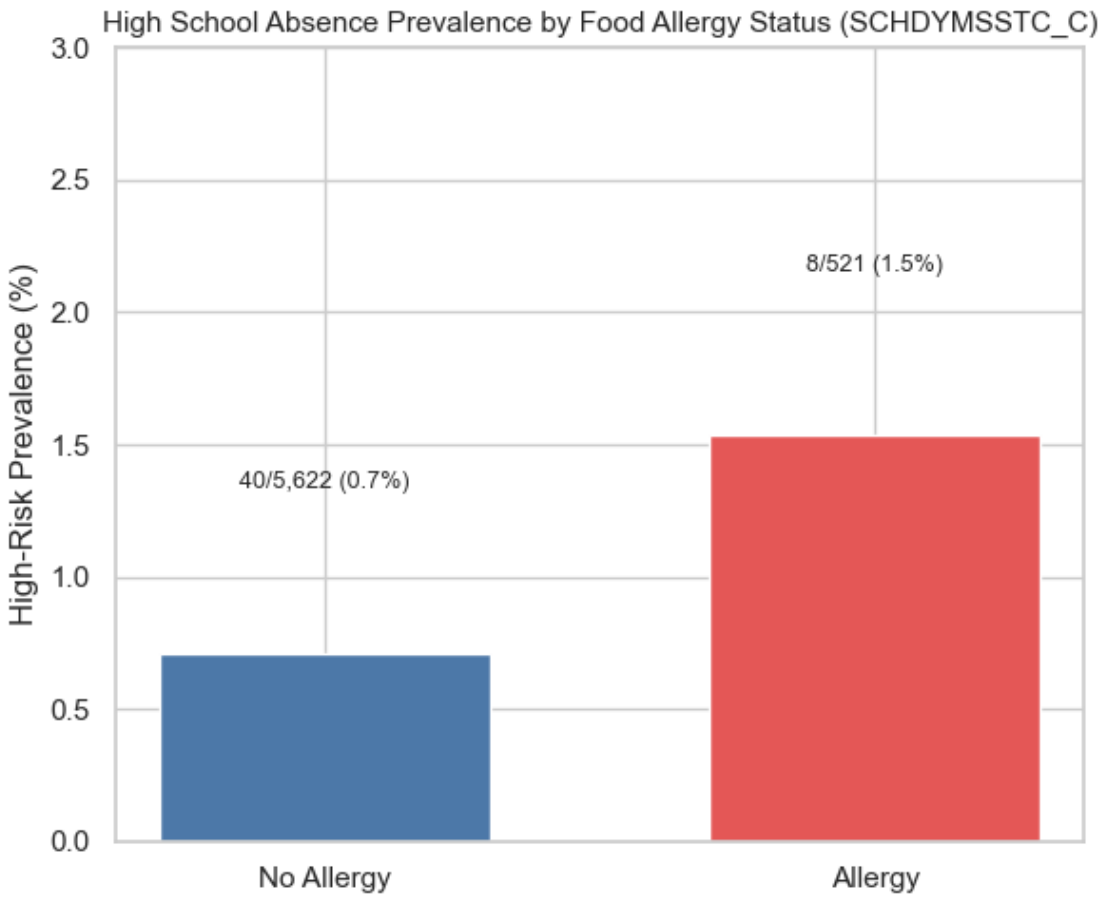


1 Analysis

What factors are associated with health status among children with food allergies? How does this differ when compared to children without food allergies?


High Anxiety Prevalence by Food Allergy Status (ANXFREQ_C)
High-Risk Prevalence (%)
30
25
20
15
10
5
0
891/5,683 (15.7%)
122/523 (23.3%)
No Allergy
Allergy

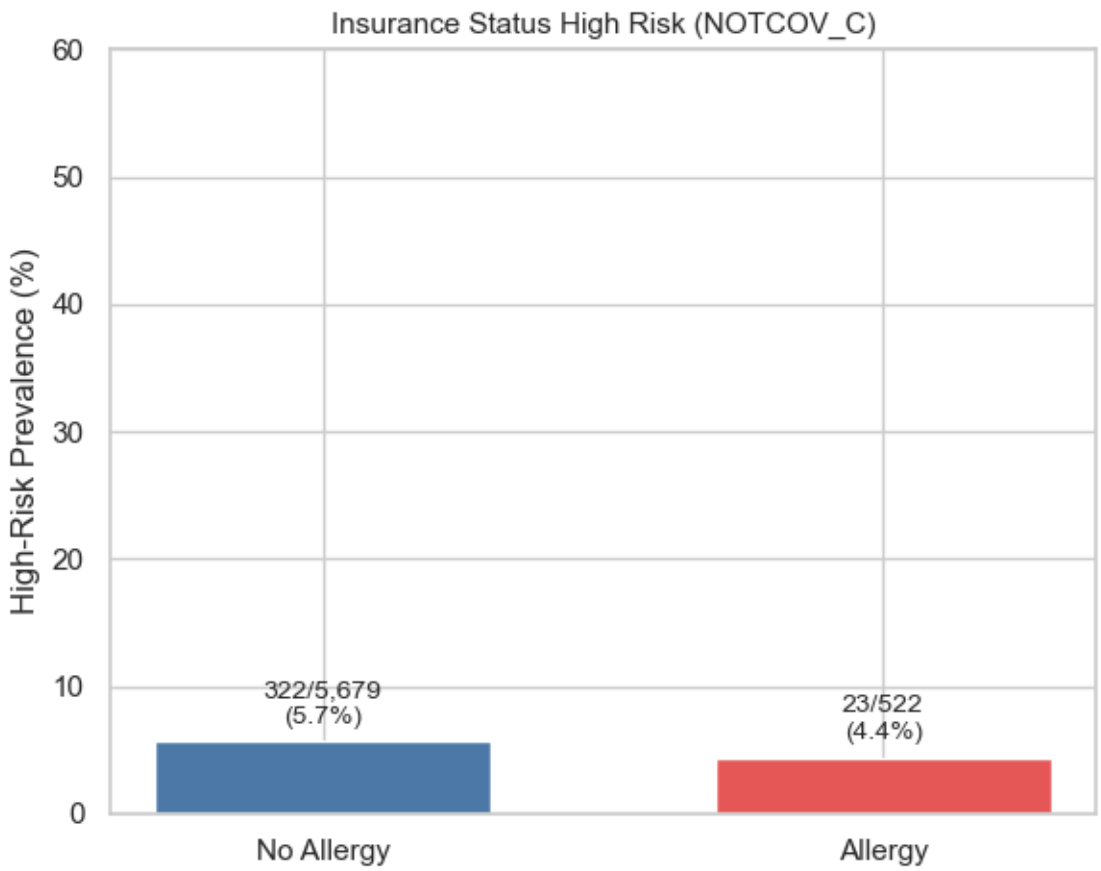

High Depression Prevalence by Food Allergy Status (DEPFREQ_C)
High-Risk Prevalence (%)
0
2
4
6
8
10
12
14
240/5,686 (4.2%)
49/523 (9.4%)
No Allergy
Allergy

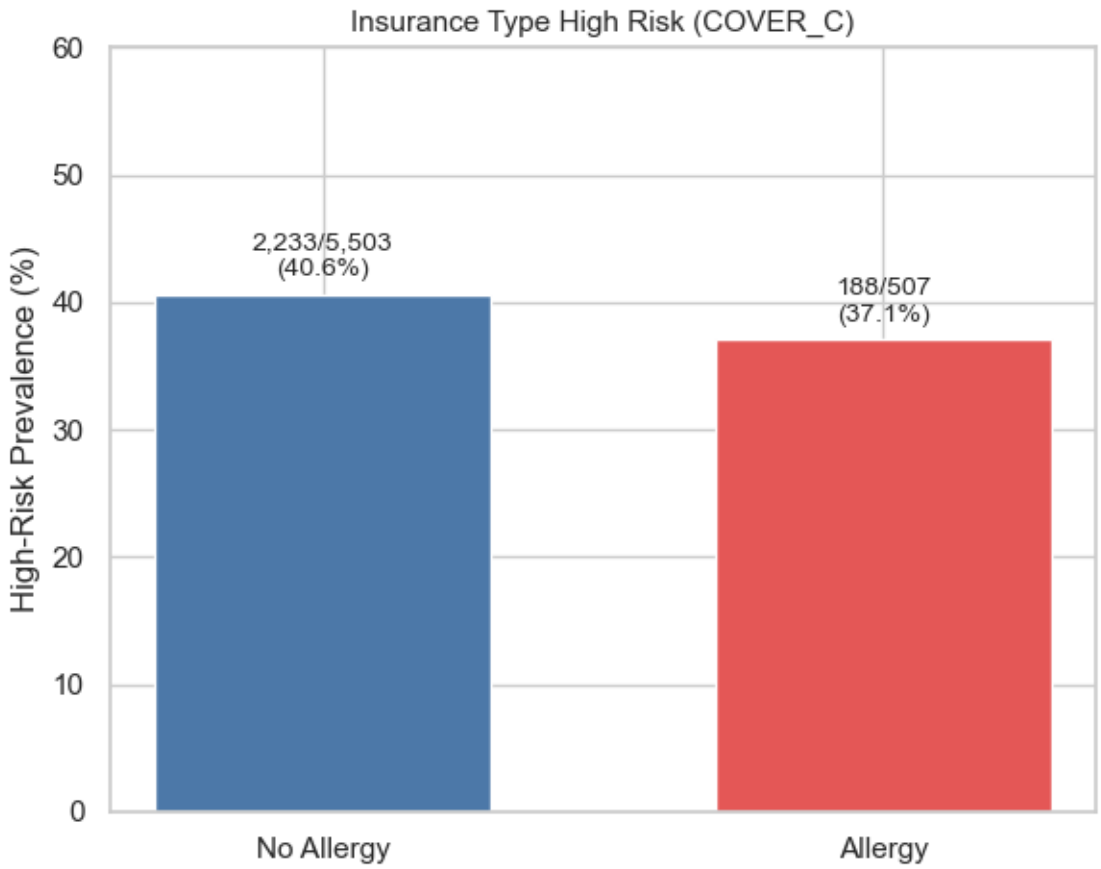


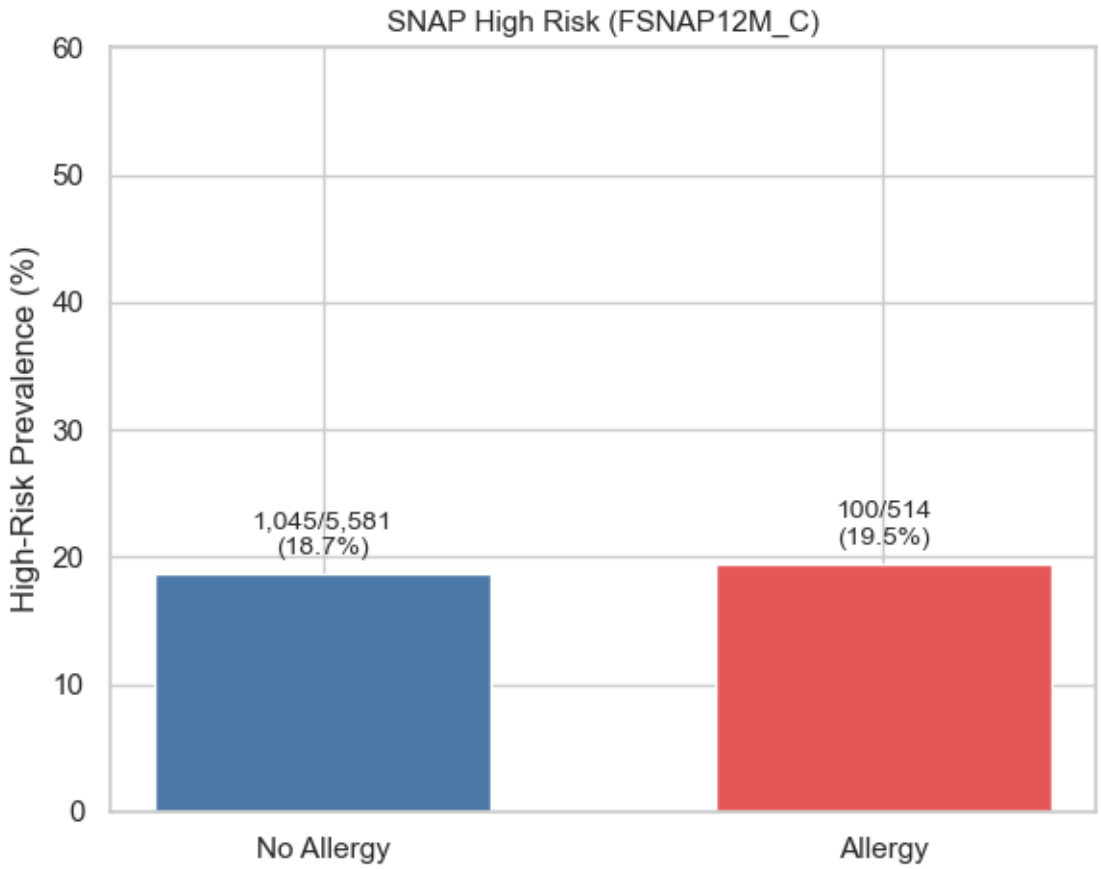

```
=== high_anxiety ===

high_anxiety  Not High-Risk  High-Risk
food_allergy
No Allergy             4792        891
Allergy                 401        122

chi2 = 19.9569, p-value = 7.92095e-06, dof = 1

=== high_depression ===

high_depression  Not High-Risk  High-Risk
food_allergy
No Allergy                5446        240
Allergy                    474         49

chi2 = 27.4546, p-value = 1.60825e-07, dof = 1

=== high_absence ===
```

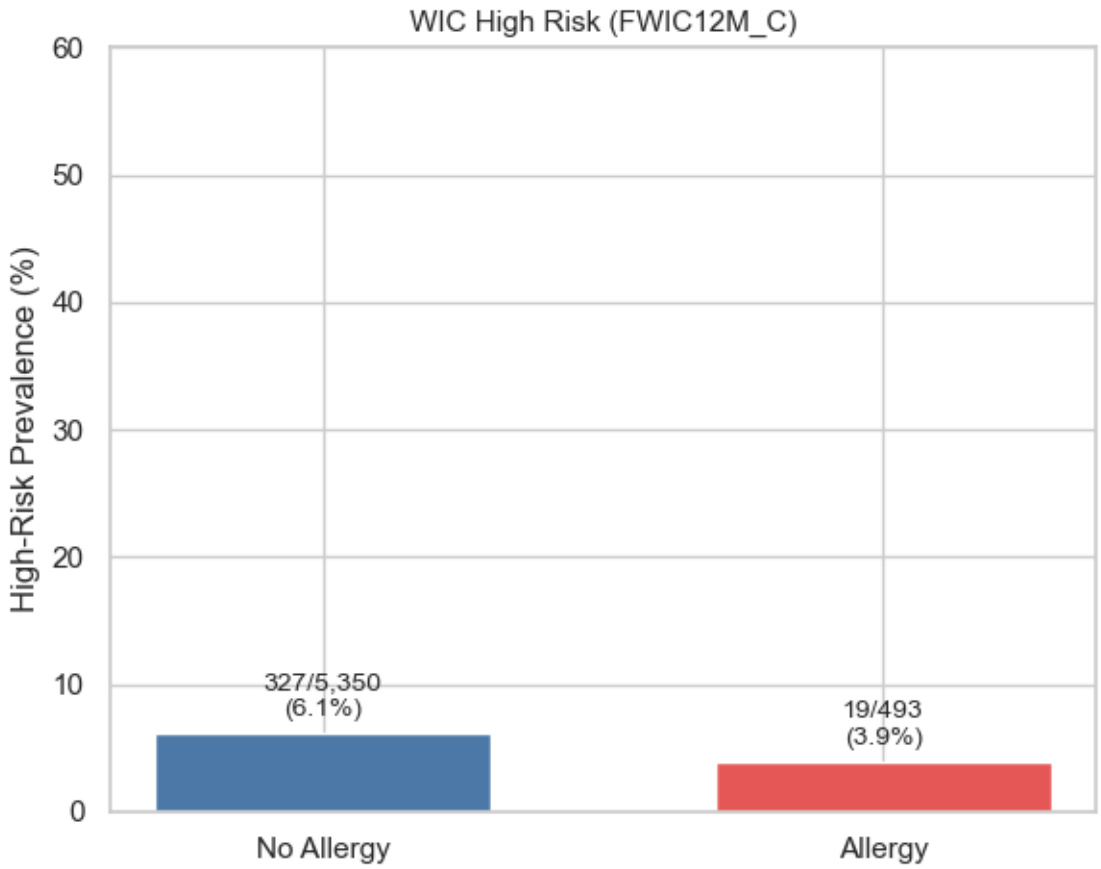
```
high_absence  Not High-Risk  High-Risk
food_allergy
No Allergy             5582         40
Allergy                 513          8
```

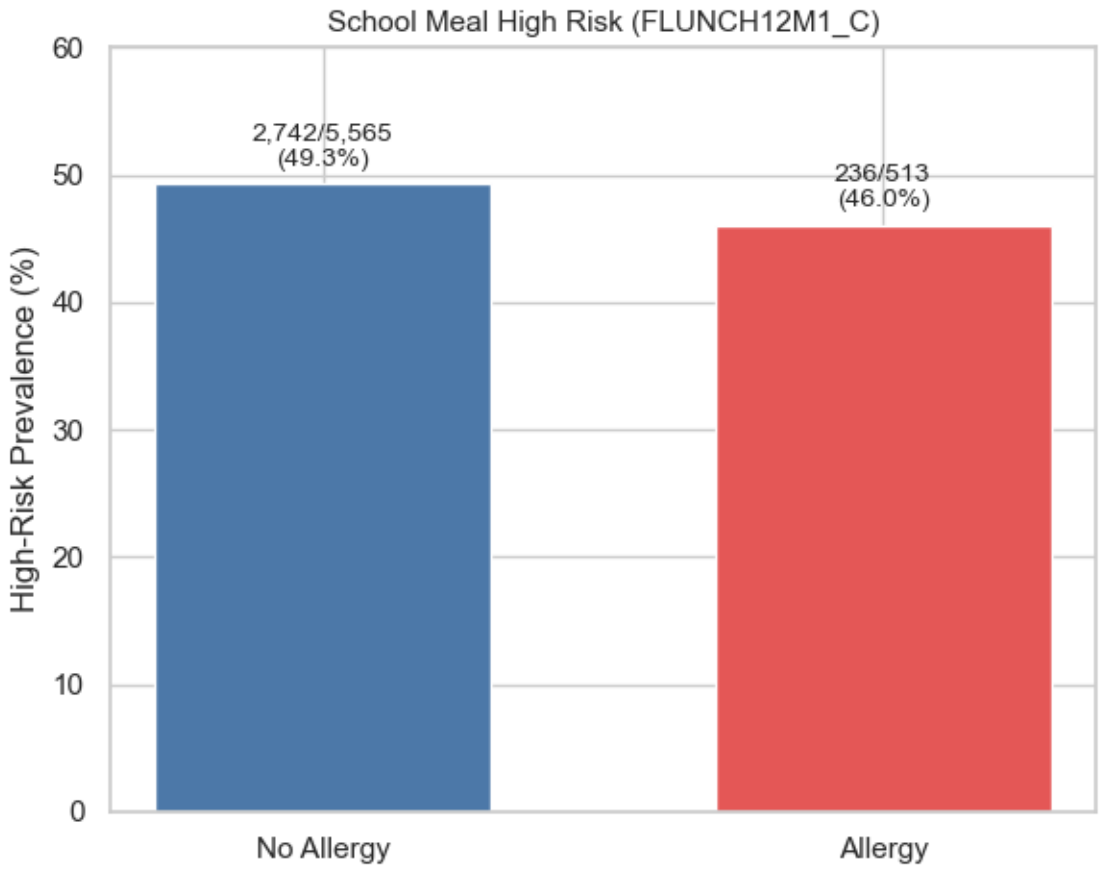
```
chi2 = 3.1808, p-value = 0.0745073, dof = 1
```



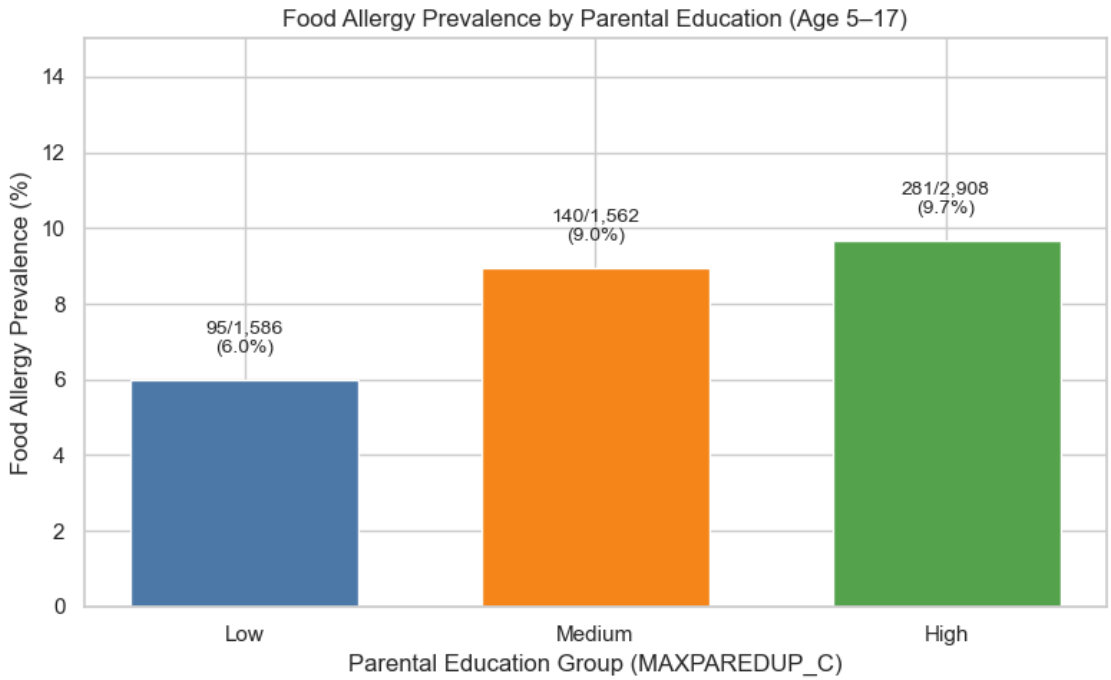

Insurance Type High Risk (COVER_C)
High-Risk Prevalence (%)
60
50
40
30
20
10
0
2,233/5,503
(40.6%)
188/507
(37.1%)
No Allergy
Allergy


SNAP High Risk (FSNAP12M_C)
High-Risk Prevalence (%)
60
50
40
30
20
10
0
1,045/5,581
(18.7%)
100/514
(19.5%)
No Allergy
Allergy

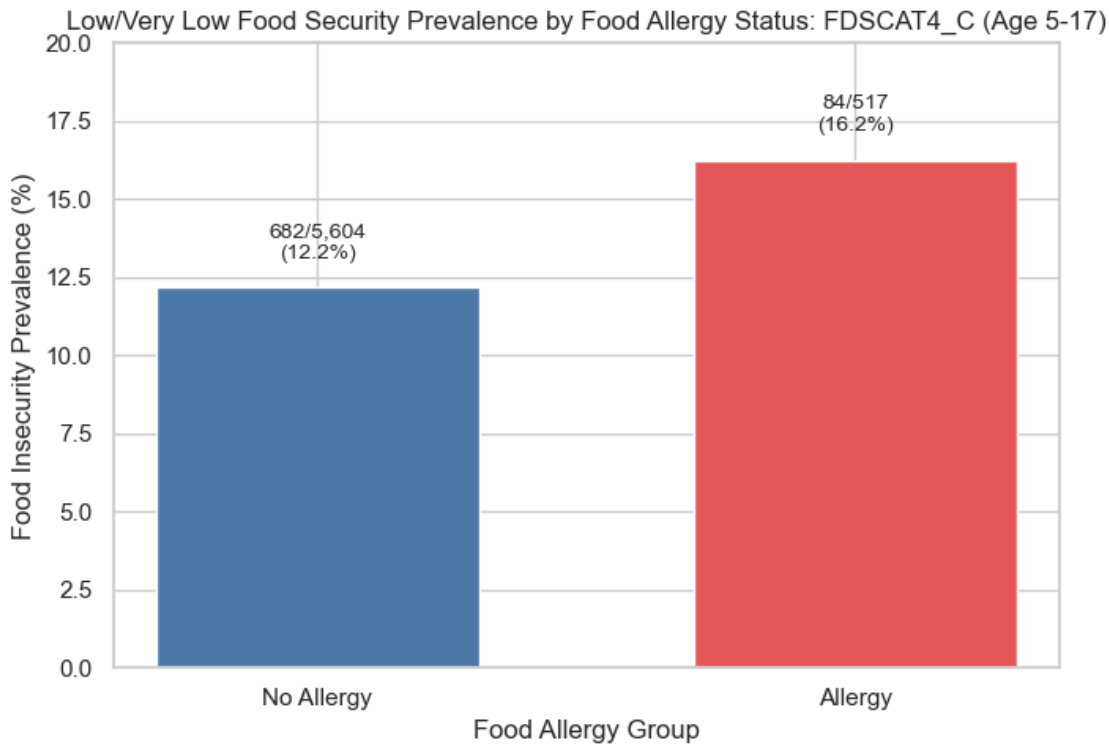

WIC High Risk (FWIC12M_C)
High-Risk Prevalence (%)
60
50
40
30
20
10
0
327/5,350
(6.1%)
19/493
(3.9%)
No Allergy
Allergy



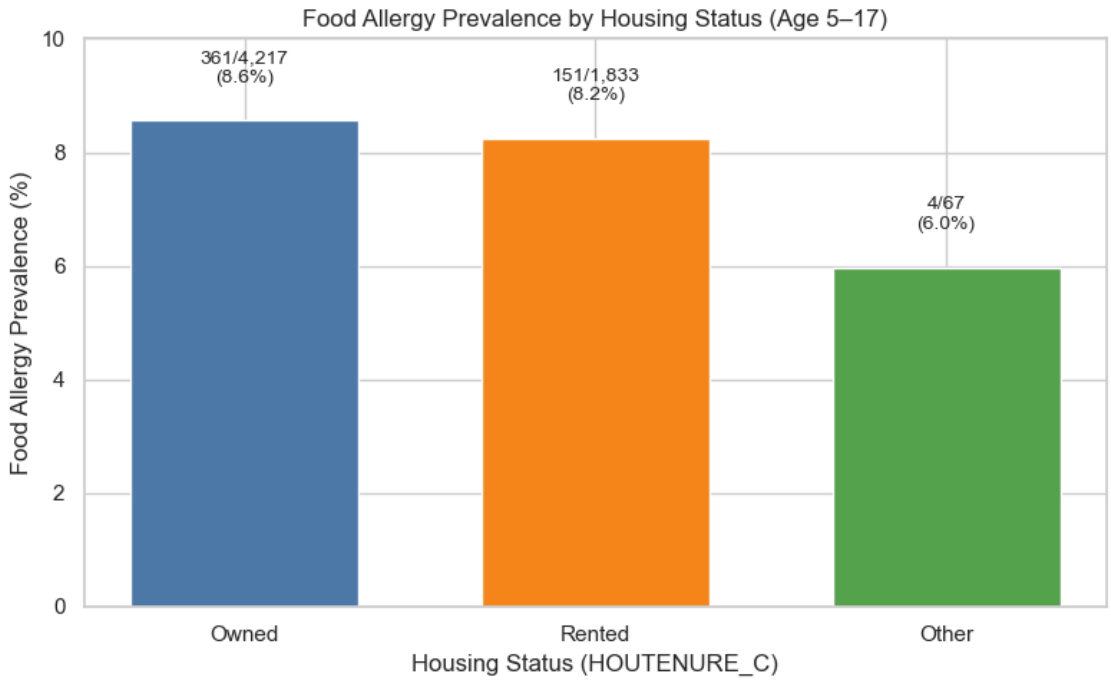

```
Crosstab: Count

  education_group  No Allergy  Allergy
0             Low        1491       95
1          Medium        1422      140
2            High        2627      281

Crosstab: Row Percentage

  education_group  No Allergy (%)  Allergy (%)
0             Low           94.01         5.99
1          Medium           91.04         8.96
2            High           90.34         9.66
```



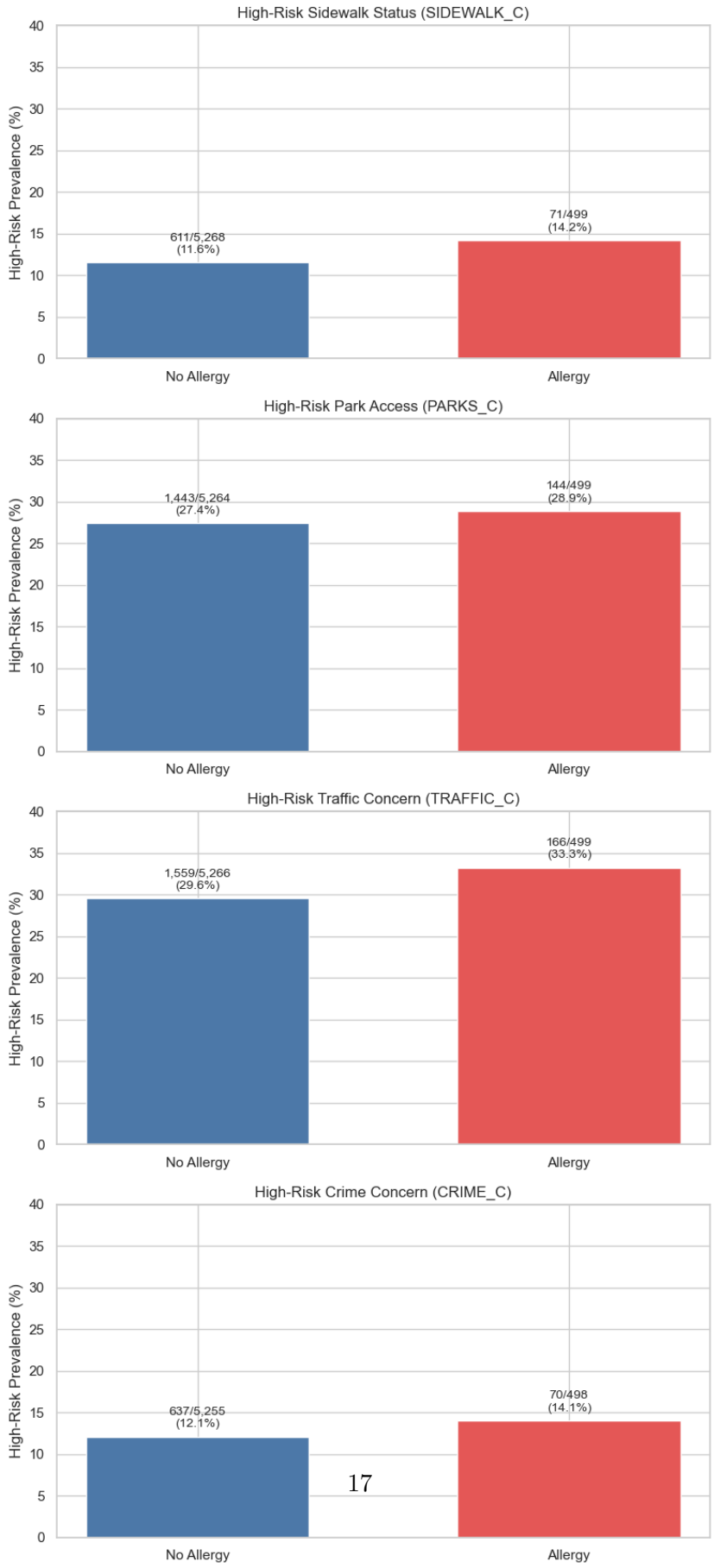

```
Crosstab (count):

food_insecure  High/Marginal  Low/Very Low
food_allergy
No Allergy              4922           682
Allergy                  433            84
```



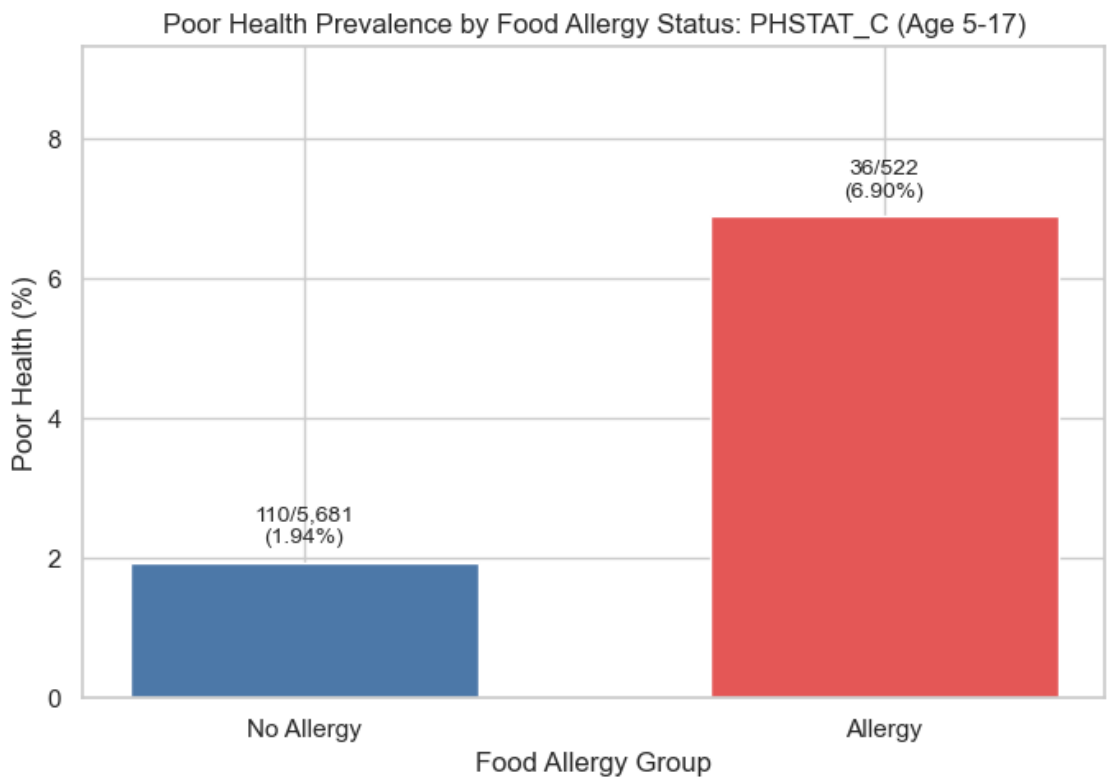

```
Chi-square test result
chi2 = 6.8211
p-value = 0.00900854
dof = 1
```


Food Allergy Prevalence by Housing Status (Age 5–17)
Food Allergy Prevalence (%)
10
8
6
4
2
0
361/4,217
(8.6%)
151/1,833
(8.2%)
4/67
(6.0%)
Owned
Rented
Other
Housing Status (HOUTENURE_C)

Built Environment High-Risk Prevalence by Food Allergy Status (Age 5-17)



```
Poor health among children with food allergy only:
36/522 (6.90%)
```




```
Crosstab (count):

poor_health   Not Poor  Poor
food_allergy
No Allergy        5571   110
Allergy            486    36

chi2 = 49.0442
p-value = 2.50256e-12
dof = 1
```